LECTURE 8

Ex

Solve $\left\{ \begin{array}{rcrcrcl} x_1 & & & + & x_3 & = & 1 \\ 2x_1 & - & x_2 & - & 2x_3 & = & 1 \end{array} \right\}$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 2 & -1 & -2 & 1 \end{array}\right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & -1 & -4 & -1 \end{array}\right] \quad (-1)R_2 \quad \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 4 & 1 \end{array}\right]$$

We have $\left\{ \begin{array}{l} x_1 + x_3 = 1 \\ x_2 + 4x_3 = 1 \end{array} \right\} \Rightarrow \begin{array}{l} x_1 = 1 - x_3 \\ x_2 = 1 - 4x_3 \end{array}$

Hence $x_3 = t$ where $t \in \mathbb{F}$

Thus $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ will be a sol'n to our system iff

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -t \\ -4t \\ t \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ -4 \\ 0 \end{bmatrix}$$

Therefore our sol'n set is

$$S = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ -4 \\ 0 \end{bmatrix} : t \in \mathbb{R} \right\}$$

Ex

$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 3 \\ 0 & 1 & -2 & 1 \\ 1 & 0 & 1 & 4 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & -1 & 1 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{array}\right] \Rightarrow \begin{array}{l} x_2 = 1 \\ x_3 = 0 \\ x_1 = 4 \end{array} \Rightarrow \left\{ \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix} \right\}$$

$$\left[\begin{array}{cccc|c} 0 & 1 & 0 & -1 & 2 \\ 1 & 2 & 1 & 1 & -1 \\ 2 & 5 & 2 & 1 & 1 \end{array}\right] \xrightarrow{\cdots} \left[\begin{array}{cccc|c} 1 & 2 & 1 & 1 & -1 \\ 0 & 1 & 0 & -1 & 2 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right] \Rightarrow 0 = 1 \Rightarrow \text{no sol'n}$$

Def'n

A <u>leading entry (pivot)</u> is the leftmost non-zero entry in a non-zero row.

leading entry (pivot)

CONTINUED

Def'n A matrix $A \in M_{m \times n}(\mathbb{F})$ is said to be in <u>row echelon form</u> (REF) if:

(1) every pivot occurs strictly to the right of the pivot above it

(2) all zero rows are below non-zero rows

Ex

$$\begin{bmatrix} 0 & 2 & 3 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ not REF} \qquad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ REF}$$

$$\left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{array}\right] \text{ not REF} \qquad \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \text{ not REF}$$

Def'n The elementary row operations (EROs) are the operations performed on the rows of a matrix that are of the form:

(1) row swap $\quad R_i \longleftrightarrow R_j$

(2) row scale $\quad cR_i \quad (c \neq 0)$

(3) row addition $\quad R_i + mR_j \quad (i \neq j,\ m \neq 0)$

We say two matrices $A, B, \in M_{m \times n}(\mathbb{F})$ are <u>row equivalent</u> if we can apply EROs to one to get the other.

$$A \longrightarrow \cdots \longrightarrow \cdots \longrightarrow B$$

<u>Gauss - Jordan algorithm (1st ed.)</u>

To solve a system of linear eqn's:

(1) Write the system in augmented matrix form $[A \mid \vec{b}]$

gauss - jordan algorithm

CONTINUED

(2) Apply EROs to $[\, A \mid \vec{b} \,]$ to turn it into a matrix in REF.

Ex Solve $\left[\begin{array}{ccc|c} 1 & 0 & -1 & 2 \\ 2 & 1 & 1 & -1 \\ 3 & 1 & 0 & 1 \end{array}\right]$.

$$\xrightarrow[R_3 - 3R_1]{R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 2 \\ 0 & 1 & 3 & -5 \\ 0 & 1 & 3 & -5 \end{array}\right] \xrightarrow{R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 2 \\ 0 & 1 & 3 & -5 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad \text{REF}$$

We have $\begin{cases} \text{free variables} : & x_3 \\ \text{pivot variables} : & x_1, x_2 \end{cases}$

Thus the sol'n set is $\left\{ \begin{bmatrix} 2 + t \\ -5 - 3t \\ t \end{bmatrix} : t \in \mathbb{R} \right\}$.

Ex Solve $\left[\begin{array}{ccc|c} 1 & 0 & -1 & 2 \\ 2 & 1 & 1 & -1 \\ 3 & 1 & 1 & 2 \end{array}\right]$

$$\xrightarrow[R_3 - 3R_1]{R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 2 \\ 0 & 1 & 3 & -5 \\ 0 & 1 & 4 & -4 \end{array}\right] \xrightarrow{R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 2 \\ 0 & 1 & 3 & -5 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$\Rightarrow x_3 = 1 \Rightarrow x_2 + 3x_3 = -5 \Rightarrow x_2 = -8$

$\Rightarrow x_1 - x_3 = 2 \Rightarrow x_1 = 3$

Thus the sol'n set is $\left\{ \begin{bmatrix} 3 \\ -8 \\ 1 \end{bmatrix} \right\}$

Note that since there are no free variables, this sol'n is <u>unique</u>.